OFFLINE EDGE AI · RASPBERRY PI · EXECUTORCH

Saral

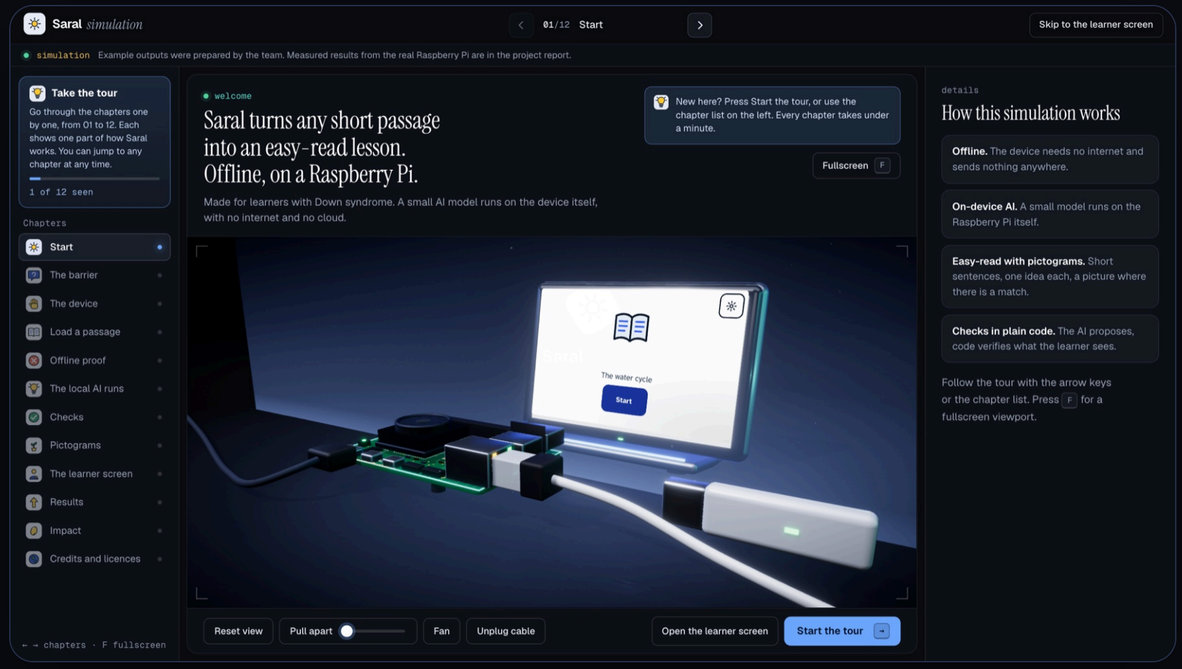
An offline study aid that turns any short passage into an easy-read lesson, with pictures and simple questions, for learners with Down syndrome.

OUR MAIN DEMO IS A WEBSITE

Explore the 3D simulation

saral-sim.pages.dev ↗

Scan or click. Opens in any modern browser. No sign-in.



The simulation: a 3D Raspberry Pi device you can orbit, pull apart and unplug.

The Primes · Kamal, Medha, Shruti · BMS Institute of Management and Technology (BMSIT&M)

This is a **simulation** of the prototype. Example outputs were prepared by the team; every metric is labelled Simulated or Measured.

THE BEST WAY TO JUDGE SARAL

Five minutes in the simulation shows the whole idea

1 **Open the link, press “Start the tour”**

12 short chapters, each under a minute. Use the arrow keys.

2 **Drag the 3D device, pull it apart**

Chapter 3. Click the Ethernet plug in chapter 5 to prove it runs offline.

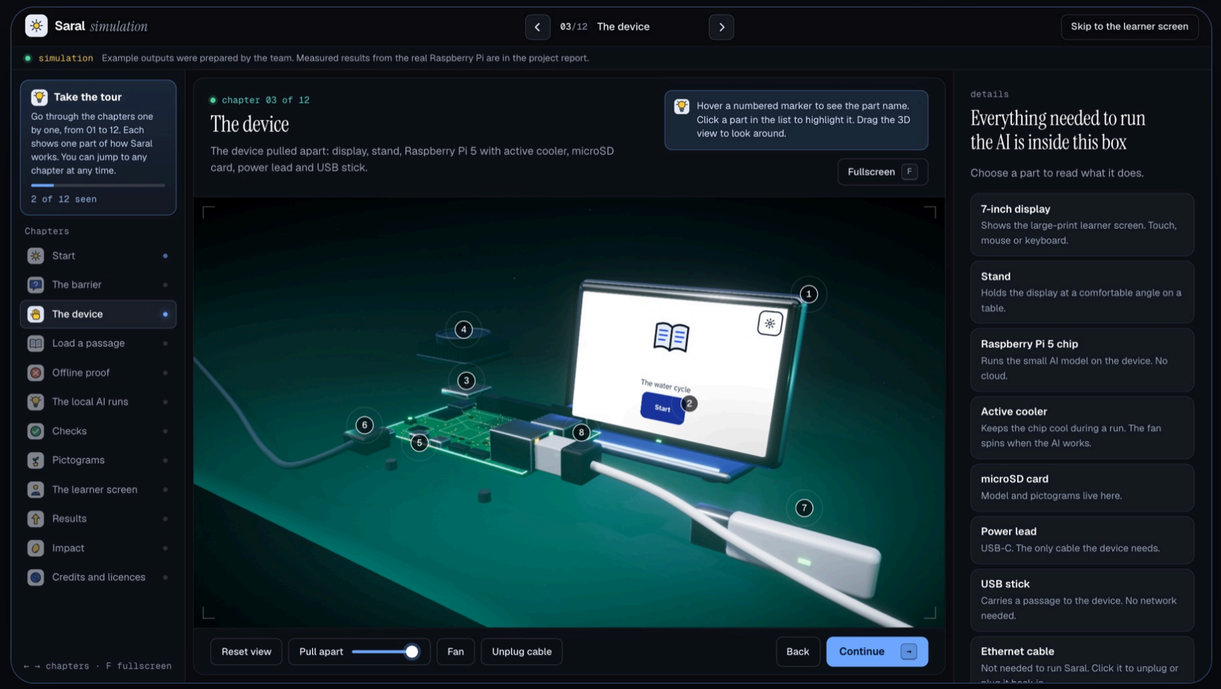
3 **Run the pipeline, watch the checks**

Chapters 6 and 7: the AI writes, plain code checks, a retry fixes it.

4 **Try the learner screen yourself**

Chapter 9: large print, one idea per screen, two big answers. You can also paste your own passage.

Open saral-sim.pages.dev ↗



Chapter 3 · the device pulled apart: display, Raspberry Pi 5, cooler, microSD, USB stick.

Everything in the site is static and offline-friendly: no accounts, no tracking, nothing loaded from other servers.

ONE CLEARLY IDENTIFIED BARRIER

Standard study text is written for readers who can follow long, abstract sentences

The barrier

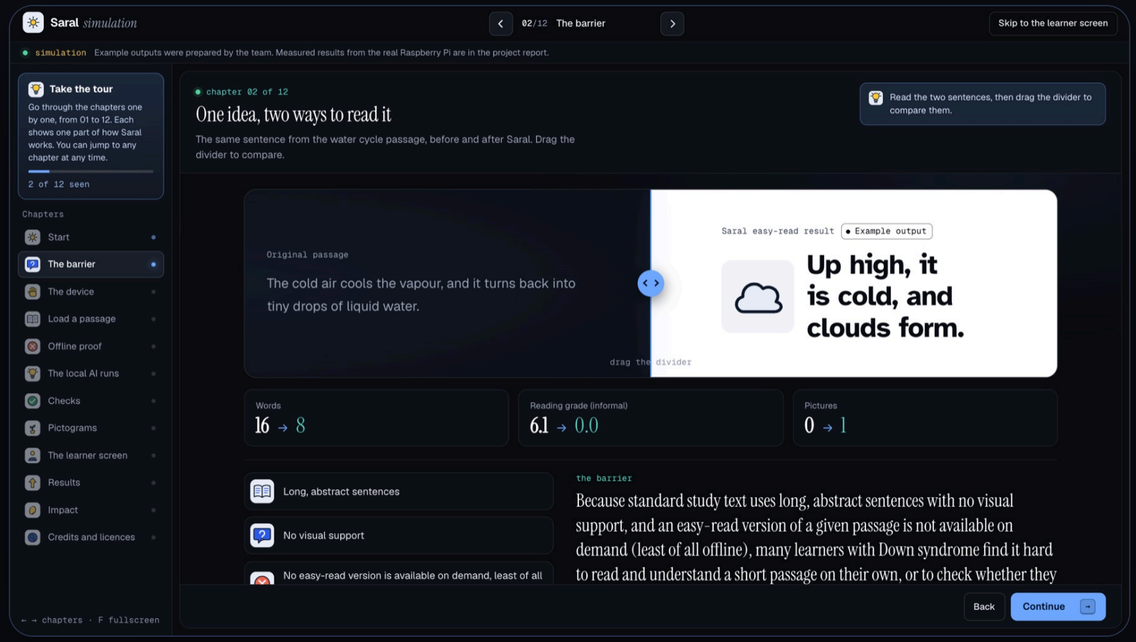
Textbook passages use long sentences, several ideas at once and hard words, with no picture support. An easy-read version of a given passage is **not available on demand**, least of all without internet.

Who it is for

Learners with Down syndrome who benefit from simplified text and visual support, using the device directly. **Learners differ widely**, so we assume no single reading level: text size and contrast are adjustable.

Why these design choices

Guidance for teachers says print should be **bigger and bolder** (Bird, SEN Magazine 80). Easy-to-Read standards ask for short sentences, one idea each (Inclusion Europe).

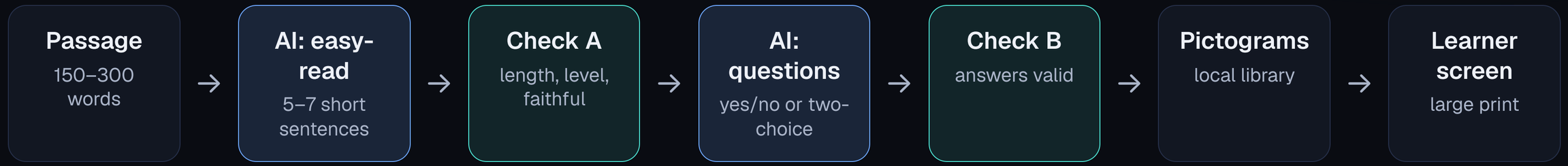


Chapter 2 · the same sentence, before (16 words, two ideas) and after (8 words, one idea, a picture).

Try it: saral-sim.pages.dev ↗

EVERYTHING RUNS ON THE RASPBERRY PI

From passage to checked easy-read lesson, offline



AI model, on device · Plain code checks

If a check fails: retry up to 2 times, then a safe fallback (selected original sentences, or a template question).

INPUT

One educational passage of about **150–300 words** (the organiser's passage on the day).

OUTPUT 1 AND 2

An easy-read summary of **5–7 sentences**, and **3 or more** yes/no or two-choice questions, each answerable from the text on screen.

OUTPUT 3

A highlighted **key word** per sentence and a **verified pictogram** where an exact match exists. No picture beats a wrong picture.

No cloud AI inference. Wi-Fi off, Ethernet unplugged, with an offline-check script as proof.

CHAPTER 7 IN THE SIMULATION

A small model can slip, so nothing reaches the learner unchecked

Check A: the easy-read text

Counts words per sentence, estimates reading level, and confirms names, numbers and key terms come from the passage.

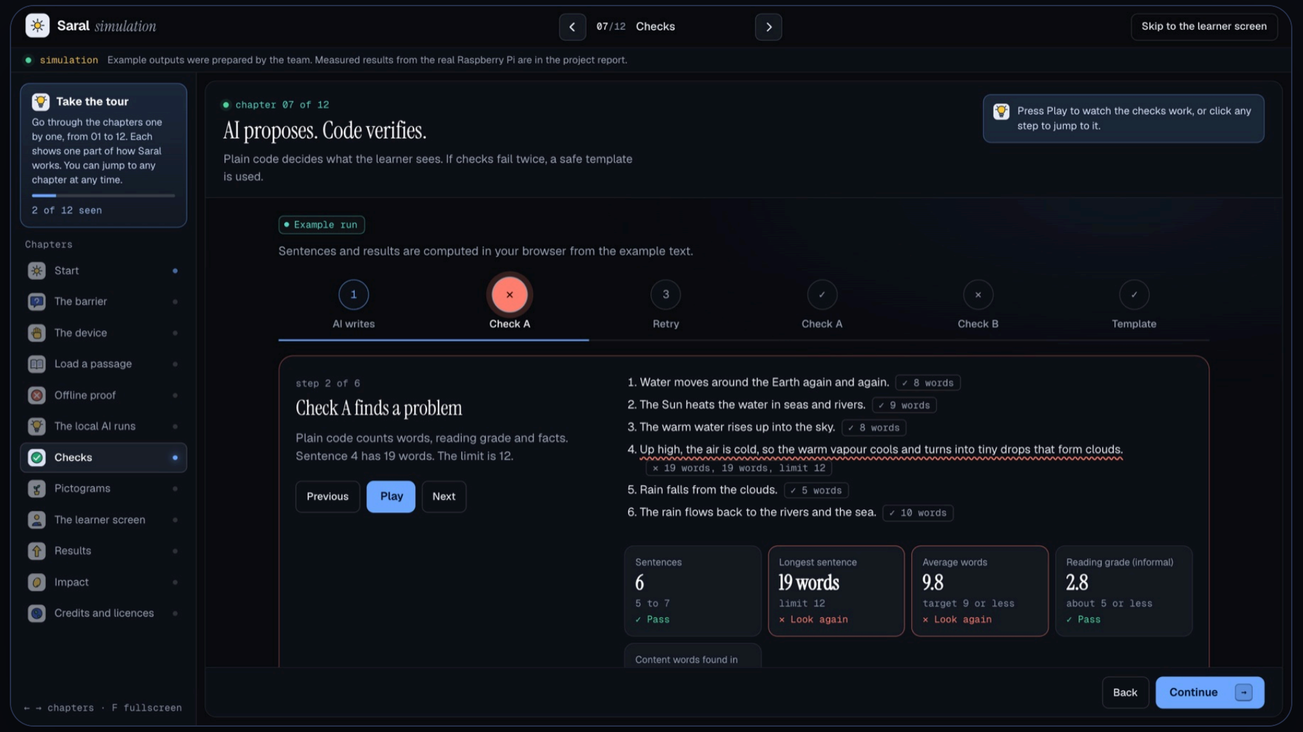
Check B: the questions

Every question must have a valid answer that can be found in the text on screen. Weak questions are dropped.

When the AI fails

Retry up to twice, then fall back to selected original sentences and template questions, so **three verified questions always appear**.

These are rough indicators, not proof of accessibility or complete factual correctness. We report failures.

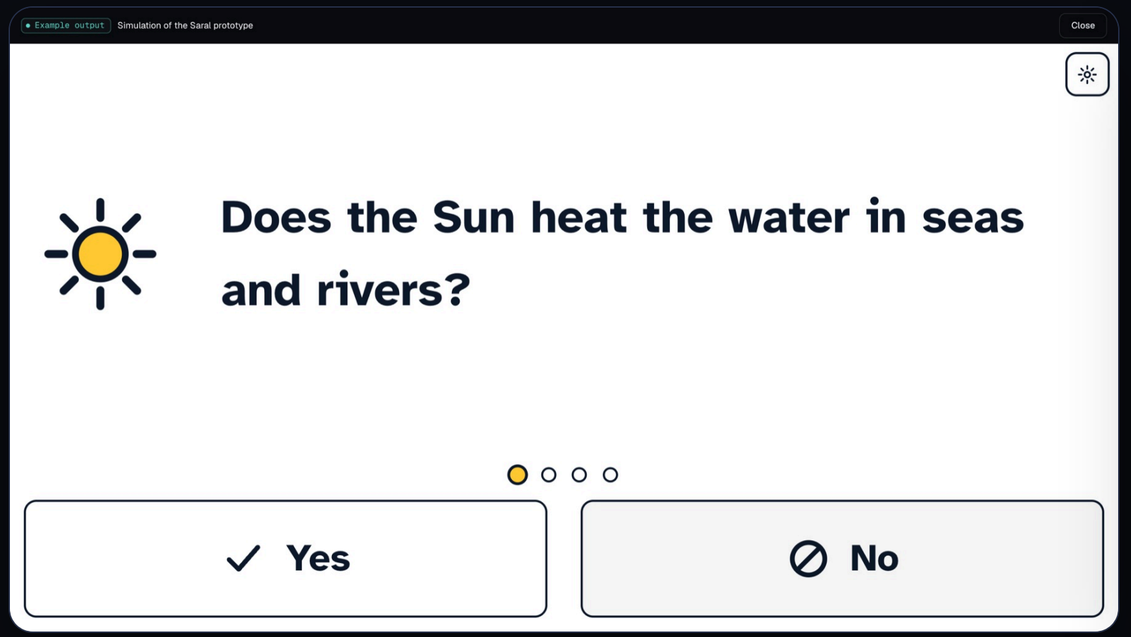


The walkthrough: Check A finds a 19-word sentence (limit 12). The retry fixes it.

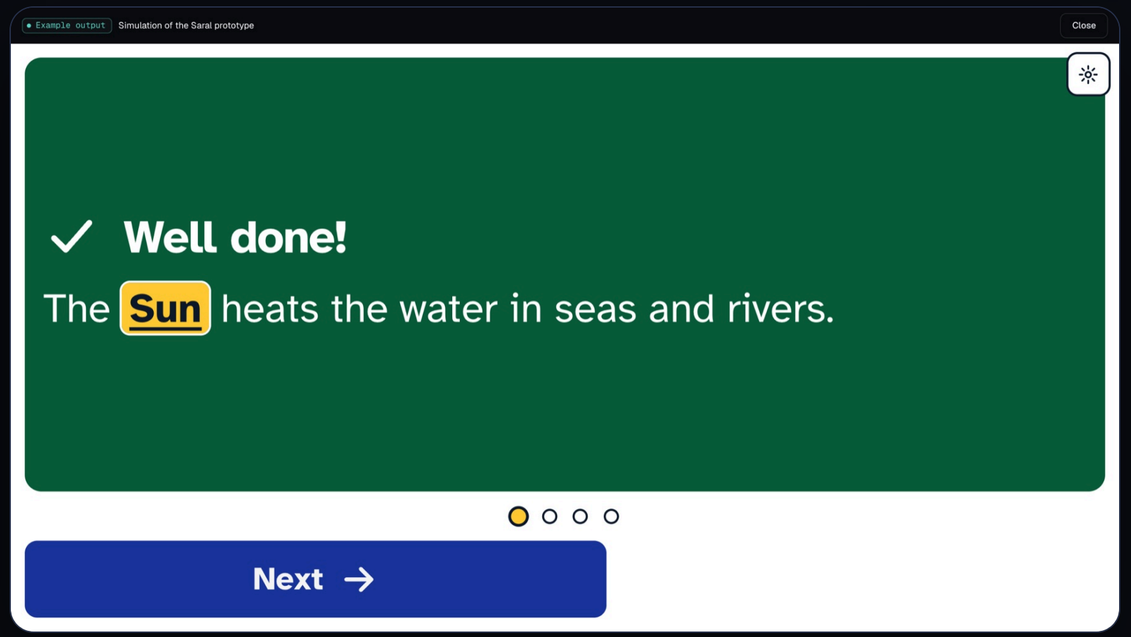
Open the Checks chapter ↗

CHAPTER 9 · THE LARGE-PRINT SCREEN

One idea per screen, two big answers, gentle feedback



A question: one picture, large type, Yes or No.



Feedback: encouraging, and it shows the supporting sentence.

- Very large, high-contrast type; seven themes
- Icons always come with words, never colour alone
- No timers, no scores, no accounts
- Keyboard and touch; a wrong answer says “Let’s look again”

Try the learner screen ↗

MAPPED TO THE PROBLEM STATEMENT

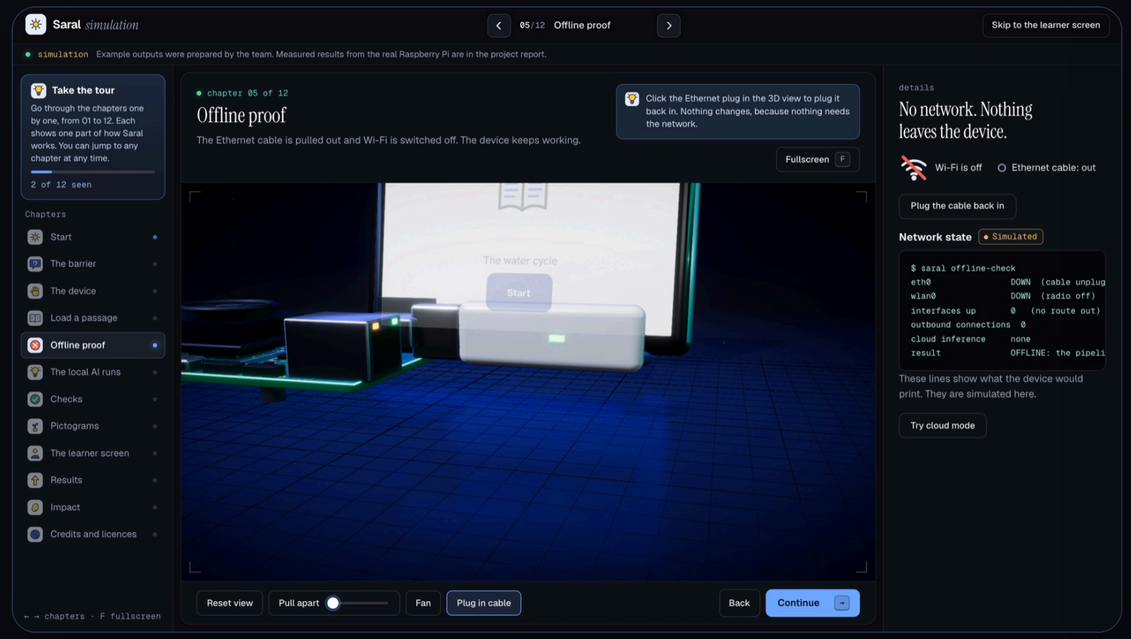
How Saral meets each of the five deliverables

ORGANISER REQUIREMENT	HOW SARAL MEETS IT	EVIDENCE ON THE DAY
1 · Barrier, learner group, how the PoC helps	Standard text is hard to read and unsupported by pictures. Learners with Down syndrome get easy-read text with pictograms on the device.	Slides 3–4; the simulation
2 · Organiser's passage, approved lightweight model, ExecuTorch on Raspberry Pi	A quantized small instruct model exported with ExecuTorch (XNNPACK). Final model is subject to organiser approval and compatibility testing.	Live run on the organiser's passage
3 · Short summary and 3+ quiz questions	5–7 sentence easy-read summary, 3–4 questions. Terminal output always works; the screen is an extra.	Terminal and large-print screen
4 · No cloud inference; time, model, size or memory	Wi-Fi off, Ethernet unplugged, offline-check script. We report time per stage, model and file size, and peak memory with and without the screen.	Network-off demo; results table measured on the Pi
5 · Impact with one example from the passage	A before/after sentence from the organiser's passage, with word counts and the picture used, and a plain statement of what we do not claim.	Impact slide filled in live

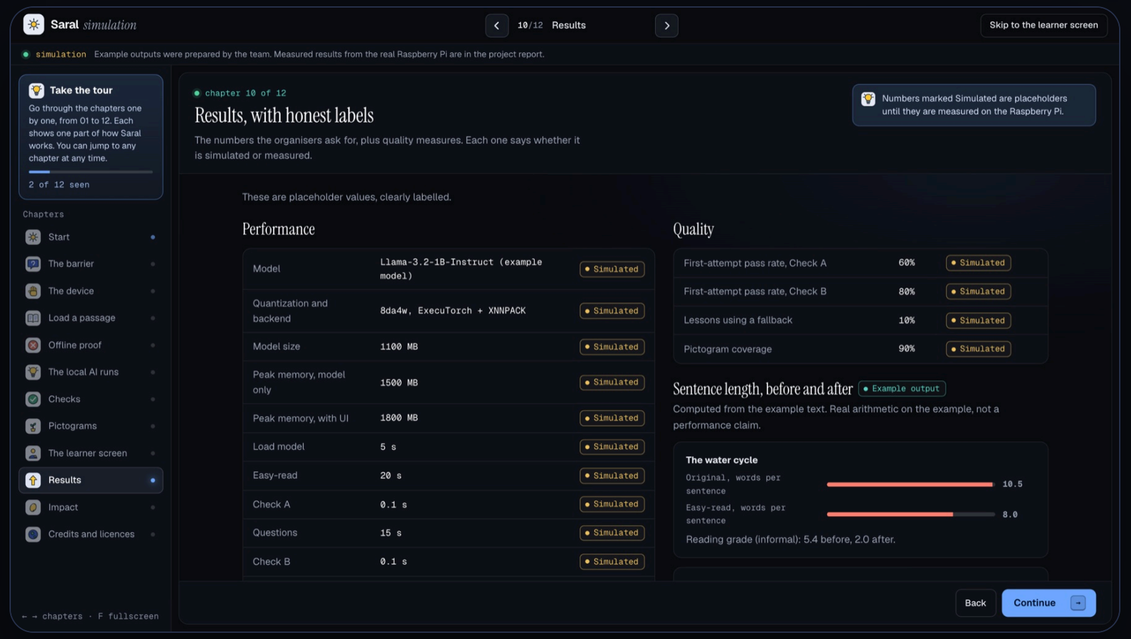
See it working: saral-sim.pages.dev ↗

HONEST NUMBERS

Offline proof, and a results table with honest labels



Chapter 5 · cable out, Wi-Fi off, the device keeps working.



Chapter 10 · every number says **Simulated** or Measured.

What we will measure on the Pi

Model, quantization, file size · peak memory (model alone, and with the screen) · time per stage and total, tokens per second · average and longest sentence, reading grade before and after · first-attempt check pass rate, fallback use, pictogram coverage.

How we measure

Real Pi, Wi-Fi off. A warm-up, then **three runs per passage**, reported as ranges. At least **five passages** from different subjects and a two-person quality review. Figures in the simulation are placeholders until then.

Open the offline proof ↗

FEASIBILITY

A small model on a Pi, with a plan for what could go wrong

Model and runtime (examples to confirm)

- **Primary:** Llama 3.2 1B Instruct
- **Backup:** Qwen3 0.6B or SmolLM2
- **Research option:** Gemma 3 1B, only if its ExecuTorch route is verified
- Quantized for XNNPACK; the exact scheme (for example 8-bit activations, 4-bit weights) is **finalised after testing**
- Choice order: organiser-approved, memory fit with the screen running, format reliability, speed

To confirm before the build

- Approved model list and Raspberry Pi model, RAM, OS
- ExecuTorch version and export requirements
- Display and other hardware; Linux dev environment

Top risks

- Broken format or long sentences: few-shot examples, targeted retries, extractive fallback
- Slow or memory-heavy: smaller model, capped tokens, terminal path always works

Still to be confirmed with the organisers: approved models, Raspberry Pi model and display.

TRUSTWORTHY ACCESSIBILITY

Careful claims, backed by what the research does and does not show

We do not claim

- **No learning-outcome claims.** We have not tested with learners.
- **Not certified Easy-to-Read.** That needs review by people with intellectual disabilities, which is planned.
- The faithfulness check works at word level and cannot catch every meaning error.

A 2026 systematic review found the evidence that Easy Read improves comprehension is limited and mostly low quality. So we claim only measurable output quality, and we plan validation with special educators.

References

- Bird, G. Helping children with Down's syndrome to learn. *SEN Magazine*, issue 80.
- Inclusion Europe. *Information for all*: European standards for making information easy to read and understand.
- UK Department of Health and Social Care. Down syndrome needs profile paper (2025).
- Wilson, H. et al. Easy Read health information for people with intellectual disabilities: a systematic review. *JARID* 39(1), 2026.
- PyTorch ExecuTorch documentation (docs.pytorch.org/executorch).

Pictograms in the simulation are original drawings. The device library will use an open set with a verified licence, credited in full.

PLAN

From proposal to the live day

By 6 Nov
Submit this proposal. Ask about approved models, the Pi and the display.

Now to 10 Nov
Pipeline, checks, pictograms and screen. ExecuTorch on a Pi as soon as we can get one.

11 to 13 Nov, if shortlisted
Finalist training, benchmarks on the real Pi, quality review, rehearsal.

14 Nov, live day, NASSCOM office
Organiser’s passage, benchmarks, offline proof, impact example, GitHub submission.

Future scope: Kannada and other languages, offline read-aloud, more subjects, review by people with Down syndrome and special educators.

TEAM

The Primes

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Contact: kamalreddi2901@gmail.com
Code: github.com/KamalReddy2901/saral-sim

saral-sim.pages.dev ↗

Please try the simulation. It is the fastest way to see Saral.